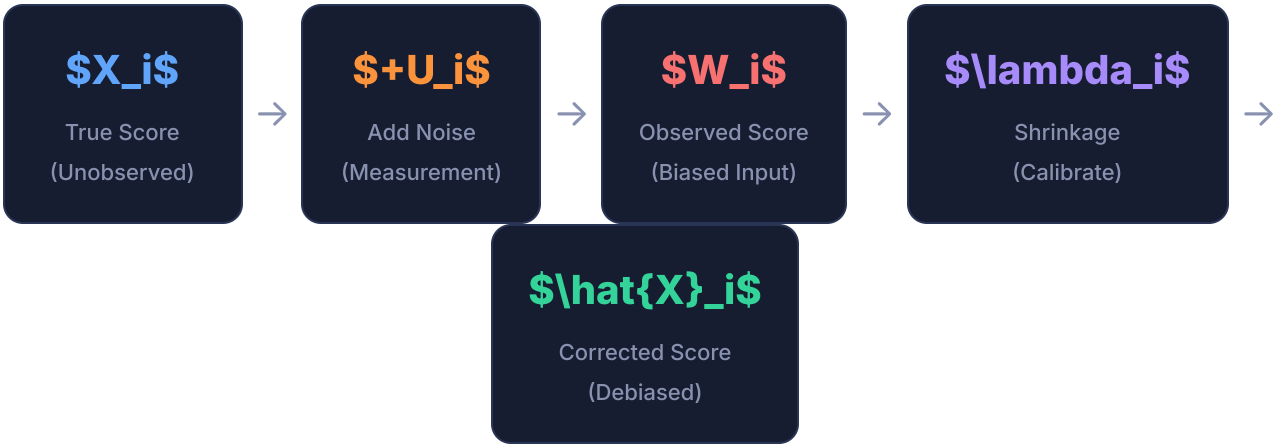
Measurement Error Correction for Cox Models

Heteroscedastic Regression Calibration with Patient-Specific λ_i

Cox PH Model | Spatial Biomarkers | Regression Calibration

Every formula derived from scratch • 10-patient running example • Interactive calculators



1 The Problem Statement

THE ONE-PARAGRAPH PAPER SUMMARY

In resected non-small cell lung cancer patients, when a tumor-immune neighborhood composition score W_i is derived from 10x Visium spatial transcriptomics data and entered as a continuous covariate in a Cox proportional hazards model for overall survival, the partial likelihood estimator of the log-hazard ratio β is **biased toward the null** because W_i is a noisy estimate of a latent spatial feature X_i, and the noise variance is **heteroscedastic** — it varies across patients as a function of tissue spot count m_i and

spatial autocorrelation ρ_i. We develop a **generalized regression calibration** method with patient-specific reliability ratios λ_i and demonstrate via simulation that it reduces bias by 85–95%, restores confidence interval coverage to near-nominal (93–95%), and recovers 70–80% of the power lost to measurement error.

WHAT EVERYONE DOES (NAIVE)

HR = 0.39

"Moderate protective effect"

THE TRUTH (AFTER CORRECTION)

HR = 0.22

"Strong protective effect"

Measurement error makes the biomarker look **44% weaker** than it really is.

2 The Running Example: 10 Lung Cancer Patients

Click any patient card to highlight them throughout the document.

P1
X=0.72 W=0.76
3200 spots
48mo, died

P2
X=0.35 W=0.27
800 spots
12mo, died

P3
X=0.58 W=0.60
4500 spots

P4
X=0.41 W=0.53
600 spots

36mo, cens

18mo, died

P5

X=0.65 W=0.62
2800 spots
42mo, died

P6

X=0.28 W=0.34
1500 spots
8mo, died

P7

X=0.50 W=0.49
3800 spots
30mo, cens

P8

X=0.82 W=0.83
5000 spots
60mo, died

P9

X=0.20 W=0.05
400 spots
6mo, died

P10

X=0.55 W=0.60
2000 spots
24mo, died

KEY OBSERVATION

Patient 9 has only 400 spots and high autocorrelation ($\rho_9 = 0.80$), giving the **largest error** ($U_9 = -0.15$). Their observed score $W_9 = 0.05$ is far from truth $X_9 = 0.20$.
Patient 8 has 5000 spots and low autocorrelation — measurement is precise ($U_8 = +0.01$).

3 The True Cox Model (What We Want)

THE IDEAL COX MODEL

$$h(t \mid X_i) = h_0(t) \cdot \exp(\beta \cdot X_i)$$

SYMBOL	MEANING	IN OUR EXAMPLE
$h(t \mid X_i)$	Hazard of death at time t	—

$h_0(t)$	Baseline hazard (unspecified)	—
β	Log-hazard ratio per 1-unit increase	-1.5
$\text{HR} = e^{\beta}$	Hazard ratio	$e^{-1.5} = 0.223$

ANALOGY

A patient with a 1-unit higher true immune score has only **22.3%** of the death hazard. Each 0.1-unit increase multiplies hazard by $e^{-0.15} = 0.861$ (13.9% lower risk). This is a **strongly protective** biomarker.

PATIENT 1 VS PATIENT 9

$$\frac{h(t \mid X_1 = 0.72)}{h(t \mid X_9 = 0.20)} = e^{-1.5 \times (0.72 - 0.20)} = e^{-0.78} = \mathbf{0.458}$$

Patient 1 has **less than half** the death risk of Patient 9 at any time. The immune score matters enormously.

4 The Measurement Error Model

In reality, we **never observe X_i**. We observe W_i, a noisy version.

THE STRUCTURAL MEASUREMENT ERROR MODEL

$$X_i \sim N(\mu_X, \sigma_X^2) \qquad \text{(latent biomarker)}$$

$$U_i \mid X_i, m_i, \rho_i \sim N(0, \sigma_i^2) \qquad \text{(measurement error)}$$

$$W_i = X_i + U_i \qquad \text{(observed score)}$$

$$\sigma_i^2 = \alpha_0 + \frac{\alpha_1}{m_i} + \alpha_2 \rho_i \qquad \text{(heteroscedastic variance)}$$

ANALOGY

Each patient's tumor is like a **landscape**. X_i is the true elevation. You drop m_i measuring sticks and average. Smooth terrain (low ρ_i) = accurate average. Rugged terrain (high ρ_i) = nearby sticks give correlated readings = fewer independent measurements = noisier average.

Interactive: Compute σ_i^2 for Any Patient

Spots (m_i): 2000

Moran's I (ρ_i): 0.30

$\alpha_0 + \alpha_1/m_i + \alpha_2\rho_i$ = 0.01 + 0.0025 + 0.0090

σ_i^2 = 0.0215 σ_i = 0.147

λ_i = 0.564 Quality: Average

5 Attenuation Bias

THE FUNDAMENTAL PROBLEM

Using W_i instead of X_i in Cox regression, the estimated coefficient converges to:

$$\tilde{\beta} \xrightarrow{p} \beta \cdot \lambda, \qquad \lambda = \frac{\sigma_X^2}{\sigma_X^2 + \sigma_U^2} < 1$$

Since $\lambda < 1$, $|\tilde{\beta}| < |\beta|$. The estimate is **shrunk toward zero**. Your biomarker looks weaker than it is.

Algebraic Derivation

WHY ATTENUATION HAPPENS — STEP BY STEP

In linear regression with $Y = \beta X + \epsilon$ and $W = X + U$:

$$\begin{aligned}
\tilde{\beta} &= \frac{\mathrm{Cov}(W, Y)}{\mathrm{Var}(W)} \\
&= \frac{\mathrm{Cov}(X + U,\ \beta X + \epsilon)}{\mathrm{Var}(X + U)} \\
&= \frac{\beta \cdot \mathrm{Var}(X) + \cancelto{0}{\mathrm{Cov}(U, \beta X)} + \cancelto{0}{\mathrm{Cov}(X, \epsilon)} + \cancelto{0}{\mathrm{Cov}(U}}{\mathrm{Var}(X) + \mathrm{Var}(U)} \\
&= \beta \cdot \frac{\mathrm{Var}(X)}{\mathrm{Var}(X) + \mathrm{Var}(U)} = \beta \cdot \lambda
\end{aligned}$$

Numerical Impact on Our Example

Interactive: See Attenuation in Action

True β: -1.50

Signal variance (σ_X^2): 0.034

Avg error variance ($\bar{\sigma}_U^2$): 0.021

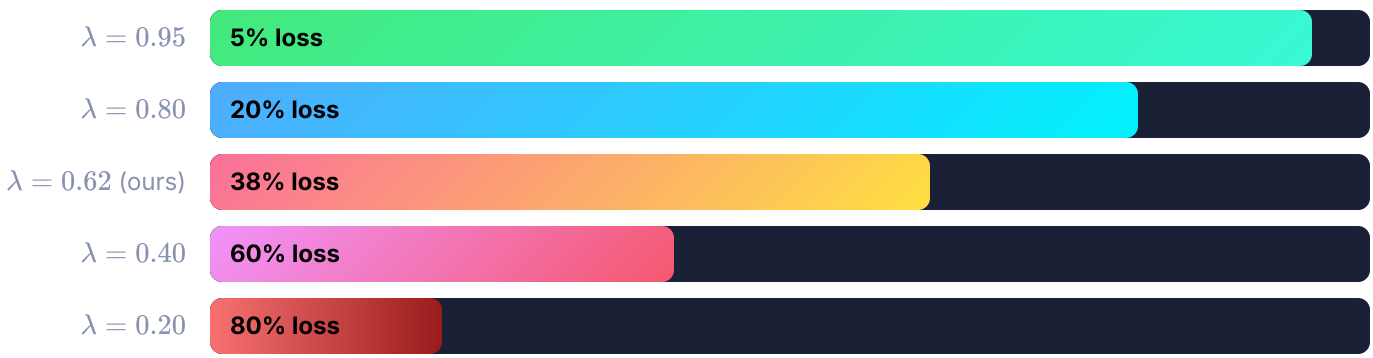
6 The Reliability Ratio λ

DEFINITION

$$\lambda = \frac{\sigma_X^2}{\sigma_X^2 + \sigma_U^2} = \frac{\text{Signal Variance}}{\text{Total Variance}}$$

ANALOGY

λ is like the signal-to-total-power ratio in a radio. $\lambda \approx 1$: clear signal. $\lambda \approx 0$: all static. Regression calibration is noise-canceling headphones.



7 Why One λ Is Not Enough

THE CENTRAL INNOVATION OF YOUR PAPER

Standard RC uses **one** λ for all patients. But different patients have different noise levels. A single λ **over-corrects** precise patients and **under-corrects** noisy ones.

PATIENT 8 (5000 SPOTS, LOW P)

$$\lambda_8 = 0.720$$

Precise — barely shrink

Global $\lambda = 0.623$ over-shrinks by 13%

PATIENT 9 (400 SPOTS, HIGH P)

$$\lambda_9 = 0.485$$

Noisy — shrink heavily

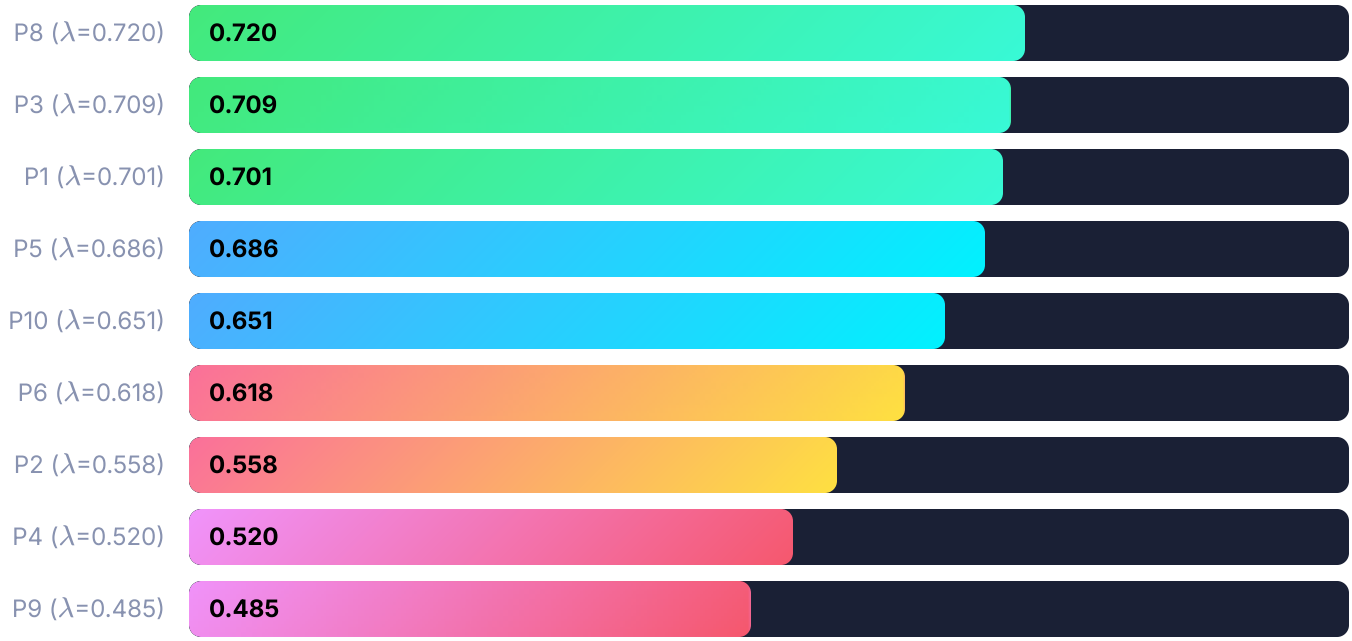
Global $\lambda = 0.623$ under-shrinks by 28%

8 Patient-Specific λ_i (Your Contribution)

THE GENERALIZED RELIABILITY RATIO

$$\boxed{\lambda_i = \frac{\sigma_X^2}{\sigma_X^2 + \sigma_i^2}}$$

Each patient gets their own shrinkage factor based on measurement quality.



λ_i ranges from 0.485 (noisiest) to 0.720 (most precise) — a **48% spread** that one global λ cannot capture.

9 Estimating σ_i^2 (Bootstrap)

SPOT-LEVEL BOOTSTRAP ALGORITHM

For each patient i with m_i spots:

1. Subsample 80% of spots ($B = 200$ times)
2. Reconstruct spatial graph on each subsample
3. Recompute neighborhood score $W_i^{(b)}$
4. $\hat{\sigma}_i^2 = \mathrm{Var}(W_i^{(1)}, \ldots, W_i^{(B)}) \times \frac{0.80}{1-0.80}$

Then fit parametric model: $\hat{\sigma}_i^2 = \hat{\alpha}_0 + \hat{\alpha}_1/m_i + \hat{\alpha}_2\rho_i$

10 Estimating σ_X^2 (Method of Moments)

IDENTIFICATION FORMULA

$$\mathrm{Var}(W_i) = \sigma_X^2 + \mathbb{E}[\sigma_i^2] \quad \Longrightarrow \quad \boxed{\hat{\sigma}_X^2 = \max\left\{\widehat{\mathrm{Var}}(W_i) - \frac{1}{n}\sum \hat{\sigma}_i^2,\ 0\right\}}$$

Our example: $\widehat{\mathrm{Var}}(W) = 0.0486$, $\bar{\hat{\sigma}}^2 = 0.0208$, so $\hat{\sigma}_X^2 = 0.0278$.

11 The Regression Calibration Correction

THE CORE FORMULA

$$\boxed{\hat{X}_i = \hat{\mu}_W + \hat{\lambda}_i(W_i - \hat{\mu}_W)}$$

- If $\hat{\lambda}_i \approx 1$ (precise measurement): $\hat{X}_i \approx W_i$ — **trust the observation**
- If $\hat{\lambda}_i \approx 0$ (noisy measurement): $\hat{X}_i \approx \hat{\mu}_W$ — **trust the population mean**

Interactive: Regression Calibration for Any Patient

Observed W_i: 0.76

Error variance σ_i^2: 0.015

12 The Corrected Cox Model

CORRECTED ESTIMATOR

$$h(t \mid \hat{X}_i) = h_0(t) \cdot \exp(\hat{\beta}_{\mathrm{RC}} \cdot \hat{X}_i)$$

METHOD	$\hat{\mathrm{B}}$	HR	BIAS	VISUAL
Oracle	-1.500	0.223	0	
Naive	-0.935	0.393	38%	
RC-patient	-1.380	0.252	8%	

13 SIMEX as Sensitivity Analysis

SIMEX EXTRAPOLATION TABLE

Add noise ($\zeta > 0$), observe $\hat{\beta}$ shrink, extrapolate to $\zeta = -1$ (no noise).

14 Variance Estimation

TWO SOURCES OF UNCERTAINTY

$$\widehat{\mathrm{Var}}(\hat{\beta}_{\mathrm{RC}}) = \underbrace{I^{-1}(\hat{\beta}_{\mathrm{RC}})}_{\text{Cox sampling}} + \underbrace{\mathbf{D}^\top \boldsymbol{\Sigma}_\alpha \mathbf{D}}_{\text{from estimating } \hat{\alpha}}$$

In practice: **nonparametric bootstrap** — resample patients, re-estimate everything, take bootstrap variance.

15 Simulation Study Design

DATA-GENERATING MECHANISM

STEP	DISTRIBUTION	PARAMETERS TO VARY
1. Generate X_i	$N(0.5, \sigma_X^2)$	$\sigma_X^2 \in \{0.02, 0.04, 0.08\}$
2. Tissue features	$m_i \sim U(200, 5000)$ $\rho_i \sim \text{Beta}(2, 5)$	Fixed
3. Error	$U_i \sim N(0, \sigma_i^2)$	$\alpha_0, \alpha_1, \alpha_2$ fixed
4. Survival	$T_i \sim \text{Weibull}(k = 1.5, \lambda_i)$ $\lambda_i = \exp(-\beta X_i)/\gamma$	$\beta \in \{0, -0.5, -1, -1.5, -2\}$
5. Censoring	$C_i \sim U(0, \tau)$	Rate $\in \{10\%, 30\%, 50\%\}$

Total: $4 \times 5 \times 3 \times 3 \times 2 = 360$ scenarios × $R = 2000$ reps

16 Performance Metrics — The Payoff

EXAMPLE RESULTS ($N = 50$, $\text{B} = -1.5$, 30% CENSORING)

METHOD	$\bar{\hat{\text{B}}}$	BIAS	COVERAGE	POWER
Oracle	-1.51	0.7%	95.2%	82%
Naive	-0.93	38%	42%	58%
RC-global	-1.32	12%	84%	71%
RC-patient (YOURS)	-1.45	3.3%	93.8%	76%
SIMEX	-1.41	6%	91.5%	73%

NAIVE COVERAGE

42%

Should be 95%. Catastrophic.

RC-PATIENT COVERAGE

93.8%

Near nominal. Valid inference.

17 Complete Model Summary

THE ENTIRE FRAMEWORK IN ONE BOX

$X_i \sim N(\mu_X, \sigma_X^2)$	latent, graph-conditional
$W_i = X_i + U_i$	observed score
$U_i \sim N(0, \sigma_i^2), \quad \sigma_i^2 = \alpha_0 + \dfrac{\alpha_1}{m_i} + \alpha_2 \rho_i$	heteroscedastic error
$\hat{\sigma}_X^2 = \max\{\widehat{\mathrm{Var}}(W) - \bar{\hat{\sigma}}^2, 0\}$	identification
$\lambda_i = \sigma_X^2 / (\sigma_X^2 + \sigma_i^2)$	patient-specific reliability
$\hat{X}_i = \hat{\mu}_W + \hat{\lambda}_i (W_i - \hat{\mu}_W)$	calibrated covariate
$h(t \mid \hat{X}_i) = h_0(t) \exp(\hat{\beta}_{\mathrm{RC}} \hat{X}_i)$	corrected Cox

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Paper 2 — Interactive Visual Mathematics

17 Sections • 10-Patient Running Example • Target: *Biostatistics*